

Ashir Rashid

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Computer Science graduate with research and engineering experience across LLMs, agentic pipelines, and applied deep learning. Published researcher in medical AI. Built end-to-end systems spanning RAG pipelines, ML microservices, and evaluation frameworks, with hands-on experience in LangChain, Azure ML, prompt engineering, and Python automation.

EDUCATION

New York University Abu Dhabi

Abu Dhabi, UAE

B.S. in Computer Science

Graduation Date: May 2026

Relevant Courses: Algorithmic Foundations of Data Science, ML, NLP, Algorithms, Data Management and Analysis

Certifications: [AWS Certified Cloud Practitioner](#)

PUBLICATIONS

"Rethinking Evaluation of Multiple Sclerosis (MS) Lesion Segmentation Models" A. Basit, A. Rashid, M. A. Hanif, M. Shafique Accepted at *IEEE IJCNN @ WCCI 2026*

WORK EXPERIENCE

GamaLearn

Abu Dhabi, UAE

Data Scientist

May 2025 - May 2026

- Designed scalable **Airflow** ELT pipelines processing **100,000+** records for downstream analytics.
- Engineered **FastAPI** microservices backed by **SQL databases**, supporting enterprise applications.
- Implemented end-to-end ML pipelines in **Azure ML Studio**, covering assessment data ingestion through real-time inference, serving a platform API used by teachers, admins, and students.
- Automated KPI reporting with **Streamlit dashboards**, cutting report generation time from ~1 hour to **minutes**.
- Built, evaluated, and backtested psychometric models using Bayesian estimation and EM-based methods, improving accuracy by **25%** and reducing calibration error (ECE) by **30%** over the previous production model.
- Wrote complex **SQL** queries from scratch involving joins across **5+** tables, feeding data into FastAPI microservices, with iterative optimization to resolve performance issues and data duplication.
- Built a **pytest**-based endpoint **testing** suite and **Docker** deployment automation scripts adopted by the full team, catching integration errors before deployment.

eBrain Lab

Abu Dhabi, UAE

Vision Deep Learning Research Assistant

Aug 2023 - May 2025

- Enhanced energy and memory efficiency of **Deep Neural Networks** by **8.3%**, including **Large Language Models**, through implementing a state-of-the-art quantization framework, expanding on an existing open framework (HAQ).
- Optimized quantization configurations by applying **non-linear optimization methods (RL, Evolutionary Search, and DNAS)**, improving model performance by **15%** and reducing hyperparameter search time by **43%**.
- Streamlined experiments with an **MLFlow**-based **MLOps** pipeline, reducing experiment iteration time by **12%**.
- Extended **nnUNet** with **MONAI**-based integrations, implementing SegResNet and Swin-UNETR from scratch, and developed custom trainers using ContourSoftDice variants to address annotator disagreement in lesion segmentation.
- Designed a custom **evaluation pipeline** with lesion-size-stratified metrics, identifying gaps in existing MS segmentation benchmarks and contributing to a framework accepted at **IEEE IJCNN 2026**.

Gantom Lighting & Controls

California, US (Remote)

Full Stack Engineer

Jan 2022 - Jul 2023

- Built an accessible **Django** web application as part of a microservices ecosystem, increasing return visits by **28%** through iterative UX improvements with the design team.
- Designed and deployed a **Firestore** database storing **500+** products, enabling efficient data storage and retrieval.
- Ensured seamless user access by deploying the application on **Google Cloud Platform** using **Google App Engine**.
- Reduced page load time by **22%** through database optimizations, caching, and asset compression strategies.
- Drove feature prioritization by conducting customer surveys and translating feedback into actionable requirements, collaborating across teams via **Asana** and **GitHub**.

PERSONAL PROJECTS

- [Agentic Process Discovery](#) - Retrieval, NLP: Administrative workflows from unstructured communication.
- [Gmail RAG](#) - LangChain, LLMs: End-to-end RAG pipeline for Gmail data, enabling semantic search and retrieval.
- [Better RAG](#) - Elasticsearch, HPC: Using Information Extraction techniques to improve RAG retrieval accuracy.

SKILLS

- Languages: Python, JavaScript, TypeScript, C++, C, Bash, SQL
- ML/AI: PyTorch, LangChain, MLflow, nnUNet, Quantization, Elasticsearch
- Web & Backend: Django, FastAPI, React, REST APIs, PostgreSQL
- Data & Analytics: Airflow, Streamlit, Excel, Power BI, Azure ML Studio
- Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Git, Google App Engine, Firestore